First, we would like to thank all reviewers for their comments. If accepted, a version of this document, together with the covered code and models, will be added to the supplementary material.
Technical Novelty: Although the main contribution is cross-script open-set OCR capability, we do introduce two technical contributions: **TC1** This work further shows that CTC can also improve attention-based models [2], even when linguistic information from the training set does *not* match the test data. This suggests part of the limitation of the attention mechanism lies in alignment and not just linguistics. **TC2** To our knowledge, this is the first framework able to handle a combination of $\{$word, character$\} \times \{$zero-shot, open-set, few-shot$\} \times \{$CJK, Latin, Indic, Yi$\}$ tasks.
Generalization: To demonstrate the generalization capability and the capability to recognize new characters without structural side-information, we here add per the reviewer's request, many-shot and single-shot experiments with oracle characters[1].We follow existing protocols for the many-shot case, whereas the single-shot removes the first 300 classes from the training set. During inference, we use the first sample in the many-shot training set as the exemplar of each class. We get 92.30 ACR for many-shot, and 91.65 seen/48.14 unseen/83.11 overall for few-shot. *This new capability allows historians to get recognition capability by simply registering a single instance without having any prior structural knowledge or fonts.*
Performance: We would like to clarify that the accuracy looks low because we use word-level accuracy (ACR$\uparrow$) instead of CER$\downarrow$[2]. Using Yi as an example, the model has a CER of 48.67, which means it recognizes more than 50% of characters in a script it has never seen, while the word accuracy (ACR) sits at around 18%. In contrast, the GPT-5.2 and additional Gemini-3.1-flash have 145/104% CER and 0 ACR[3]. Notably, both can output Yi characters, but do not match at all. 0 ACR despite having Yi tokens also holds with the Gemma4-27B and Qwen-3.6-26B models we tried with Ollama. Note that for some samples, these two can timeout, where an empty string will be returned. We also tried a few prompts for Gemini, but none led to > 0 ACR or any sign of rejection. This indicates that although the tokens exist in both LLMs, they still demonstrate fundamental problems with handling OOD data. In addition, the performance on Yi looks worse than OmniOCR [1] because they only use 30 classes in a single character OCR setup, while we use all 1,165 classes with word-length running up to 10.
Dataset Size The size of the Yi dataset we used to benchmark our model (Table 2&3) is **5000**, 500 words for each length ranging from 1 to 10. We sampled 50 for the Ours vs LLMs test to reduce token cost, exchanged with slight ACR instability: 16.12%@5k vs 18%@50 for Ours, which we deem acceptable given that both ChatGPT and Gemini stay at 0%. We will add dataset sizes and other specifications to the paper, as this critical information was inadvertently deleted when reducing the paper to the page limit.

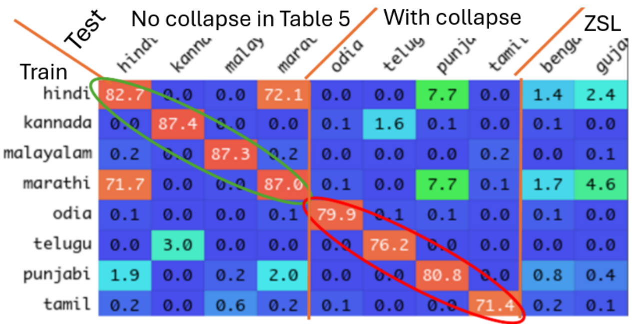

Figure 1. Affinity-Conflict analysis, numbers in ACR$\uparrow$

Script Conflict/Task Decomposition: The script conflict is more likely rooted in difficulty/training dynamic, rather than visual affinity or weight sharing. *First*, we analyzed script affinity across the Indic family by training the baseline method on only one script and testing it on itself and other scripts. We use the ACR as an indicator of affinity (Rebut Fig. 1). Results suggest the collapse is unlikely to be correlated with visual affinity, given that Malayalam does not suffer from collapse (Table 5) despite its minimal affinity with others. *Second*, we maintain the training task separation in the paper, and share modules across tasks to see if weight sharing causes collapses. We found that even though we force the 3 tasks to use the same modules, the collapse is not observed on any of the scripts. It's perfectly possible to use one non-MoE model to infer all scripts, just not for training. This shows the problem is not caused by weight-level conflicts but the training dynamic. *Finally*, to further understand which part of the training dynamic caused the problem, we enlarged the # classes sampled for each batch from 128 to 256 and 512. However, sampling more classes fails to solve the collapse when all scripts are trained together, suggesting the problem is not caused by the reduction of characters for each script in each training batch.
Notably, the collapsing scripts have lower ACR when trained only with their training set and tested on its testing set (red circle vs green in Rebut Fig 1). This give us the suspicion of the difficulty gap among scripts being the cause, and the proposed separation is a simple yet efficient alleviation. Full details and qualitative results will be added to the final paper and appendix.
Fairness We modified the WnA authors' code and trained a single expert WnA-XL at 32×128 input size on our hardware/software environment. WnA-XL-Mono yields 14.81 on Korean and 44.42 on Japanese in terms of ACR, which is lower than ours (46.30 on Japanese/22.83 on Korean).

References

[1] Liu, B., Zhang, Z., Meng, B., Wang, H., Zhang, H., Wang, C., Ergu, D., Cai, Y.: Omniocr: Generalist ocr for ethnic minority languages. arXiv:2602.21042 (2026) 1

[2] Saharia, C., Chan, W., Saxena, S., Norouzi, M.: Non-autoregressive machine translation with latent alignments. In: EMNLP. pp. 1098–1108 (2020) 1

[1] https://github.com/Pengjie-W/HUST-OBC

[2] We use ACR to avoid confusion in future comparisons, as two different CER normalizations coexist (per sample or by total GT length).

[3] CER $\geq$ 100% when PRED is longer than GT and completely wrong